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Cefazolin versus Cloxacillin or Flucloxacillin for Methicillin-Susceptible *Staphylococcus aureus* bacteremia*:* A Randomized Clinical Trial

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Abstract

Background

Whether cefazolin or an antistaphylococcal penicillin should be preferred for treating methicillin-susceptible *Staphylococcus aureus* (MSSA) bacteremia is unclear.

Methods

In an ongoing, international, Bayesian adaptive platform trial, we conducted an open-label, randomized comparison of cefazolin versus an antistaphylococcal penicillin (flucloxacillin or cloxacillin) in adult patients with penicillin-resistant, methicillin-susceptible *S. aureus* bacteremia. The primary outcome was 90-day all-cause mortality, evaluated using a hierarchical Bayesian logistic regression model. We report the posterior probability of non-inferiority (pre-specified as an adjusted odds ratio [aOR] < 1.2, approximating < 2.5% absolute difference if mortality in the (flu)cloxacillin arm was 15%) and of superiority (corresponding to an aOR <1.0). Secondary safety outcomes included the development of acute kidney injury within 14 days.

Results

This trial domain was conducted between February 17, 2022, and August 7, 2024, at which time non-inferiority was met. Death at 90 days amongst evaluable adults was 15.0% (97/645) for cefazolin and 17.0% (109/642) for (flu)cloxacillin (aOR 0.81; 95% Credible Interval [CrI], 0.59–1.12; probability of non-inferiority 99.2% and superiority 89.8%). Acute kidney injury occurred in 92/660 (13.9%) patients in the cefazolin arm versus 127/648 (19.6%) in the (flu)cloxacillin arm (aOR 0.67, 95%CrI 0.50–0.89; probability of superiority 99.7%).

Conclusion

In patients with MSSA bacteremia, cefazolin had a high probability of non-inferiority to (flu)cloxacillin for 90-day mortality, with less acute kidney injury.
(Funded by National Health and Medical Research Council and others; Clinicaltrials.gov NCT05137119).

S. aureus bacteremia is a leading cause of bacterial-related mortality worldwide[1] with more than 1 in 4 afflicted patients dying within 90 days.[2] While there are accepted best practices in the management of *S. aureus* bacteremia,[3] the preferred antibiotic for treating methicillin-susceptible *S. aureus* (MSSA) is unclear.[4] Expert opinion from endocarditis guidelines[5,6] favors the antistaphylococcal penicillins (e.g., nafcillin, oxacillin, cloxacillin, or flucloxacillin) over cefazolin[5] due to theoretical concerns about the "cefazolin inoculum effect", an *in vitro* destruction of cefazolin by beta-lactamases which may be relevant in high burden infections.[7,8] However, meta-analyses of observational studies suggest that cefazolin could be superior in terms of 30-day mortality (odds ratio 0.73; 95%CI 0.62-0.85)[9] with a more favorable adverse event profile when compared to antistaphylococcal penicillins. Given observational analyses may be subject to multiple biases,[10] a randomized clinical trial is the most reliable way to inform patient management.

We launched the *S. aureus* Network Adaptive Platform (SNAP) trial in 2022 to answer multiple questions related to the treatment of patients with *S. aureus* bacteremia.[11] Within this Bayesian adaptive platform trial, we conducted a pragmatic, open-label, randomized comparison of cefazolin to an antistaphylococcal penicillin (cloxacillin or flucloxacillin depending on the country [hereafter (flu)cloxacillin]) for the treatment of MSSA bacteremia.

Methods

Study Design and Setting

SNAP contains a series of investigator-initiated, international, multicenter studies.[11] The platform features a master protocol and integrated statistical framework[12] with domain specific appendices containing nested questions. Healthcare consumers were involved in trial design through focus groups and representation on steering committees. Each domain[11] refers to a management question with at least two interventions being compared (e.g., backbone antibiotics, adjunctive clindamycin therapy[13], or early oral switch[14]; see **Supplementary Appendix**). The backbone domain is further divided into silos based on *S. aureus* antibiotic susceptibility: penicillin-susceptible (PSSA); methicillin-susceptible, penicillin-resistant (MSSA); and methicillin-resistant (MRSA).

Herein we report the open-label, parallel group, non-inferiority, randomized comparison of patients in the MSSA silo of the antibiotic backbone domain of SNAP. Participants in this domain and silo were recruited between February 17, 2022, and June 21, 2024, at 91 sites in 8

countries (Australia, Canada, Israel, New Zealand, the Netherlands, Singapore, South Africa, and the United Kingdom; see **Supplementary Appendix)** with the last participant reaching 90-days of follow-up on September 21, 2024. Ethics and regulatory approval were obtained at each participating center. Written or oral informed consent, in accordance with regional regulations, was obtained from all patients and/or their surrogates. The study was designed by the investigators via international working groups. Data was analyzed by a dedicated analytic team of investigators. Tong, Norton, Mahar, McGlothlin, and Dymock vouch for the data and analysis. The study was supported by funding organizations and academic sponsors in multiple countries (**Supplementary Appendix**) who had no role in the design, analysis, or reporting of the results. The master protocol,[11] statistical methods,[12] domain specific appendix, and statistical analysis plan are available in the **Supplementary Appendix** and the protocol at nejm.org. SNAP is registered on clinicaltrials.gov (NCT05137119).

Participants

We enrolled hospitalized patients with *S. aureus* bacteremia to the platform within 72 hours of their index blood culture collection. This report describes adults (≥18 years). The pediatric study is ongoing. Platform and domain eligibility criteria are detailed in the **Supplementary Appendix**. Notable platform exclusions included enrollment beyond 72 hours from collection of the first positive blood culture, polymicrobial bacteremia, or patients not for active treatment. Key domain specific exclusions included an active history of penicillin or cefazolin allergy; current receipt of maintenance dialysis; and ongoing treatment with other systemic antibacterial agents with *S. aureus* activity. Additionally, because the cefazolin inoculum effect is dependent on the production of penicillinase, we only included penicillin-resistant MSSA (see *Laboratory Methods*) to avoid bias towards non-inferiority of cefazolin.

Randomization and Allocation

All participants were randomized 1:1 within the MSSA silo at platform entry by simple randomization without stratification using a web-based interactive randomization system (Spiral Software, New Zealand). As soon as participants were confirmed to be eligible for the MSSA silo of the backbone domain, their randomized allocation to cefazolin or (flu)cloxacillin was revealed.

Interventions

Recommended dosing was: 2g intravenously (IV) q8h for cefazolin (q6h for critical illness or endocarditis); 2g IV q6h for flucloxacillin (q4h for critical illness or endocarditis), and 2g IV q4h for cloxacillin, with dose adjustments for renal function (**Supplementary Appendix)**. Other than potential treatment assignment in the adjunctive antibiotic domain,[13] post-enrollment administration of additional antibiotics was discouraged.

Infectious diseases consultation was advised. The protocol recommended allocated antibiotics be continued for the entire intravenous treatment duration, with a minimum antibiotic duration of 14 days for uncomplicated bacteremia and 28–42 days for complicated bacteremia.[15] Subsequent randomization to the oral switch domain[14] was permitted at 7 and 14 days, respectively for eligible participants. Apart from the antistaphylococcal antibiotics used, clinical management was left to the treating clinician's discretion.

Outcomes

The primary outcome for the platform and domain was all-cause mortality at 90 days after platform entry. Secondary platform outcomes included: all-cause mortality at 14, 28, and 42 days; microbiologic treatment failure and diagnosis of new foci of infection between days 15 and 90; and *Clostridioides difficile* infection within 90 days. Secondary domain specific outcomes (further defined in **Supplementary Appendix**) included acute kidney injury (AKI), defined as an absolute creatinine increase ⩾26.5 μmol/L (0.3mg/dL) within 5 days or a relative increase ≥50% from baseline within the first 14 days, initiation of renal replacement therapy, and hepatotoxicity.

Adverse Events

In addition to the pre-specified safety outcomes, site investigators were asked to conform to regional requirements and, at a minimum, record serious adverse reactions defined as all serious adverse events that were thought to be possibly, probably, or definitely related to study drugs or study procedures.

Laboratory Methods

Methicillin and penicillin susceptibility were determined locally using automated testing methods and/or phenotypic disc diffusion[16] and/or molecular testing for the *mecA* and *blaZ* genes (**Supplementary Appendix)**.

Statistical Methods

Study Populations

Populations were defined based on the principle of intention-to-treat (ITT) for each estimand corresponding to the revealed randomized treatment assignment, regardless of receipt. The primary analysis was also performed on a protocol-adherent population (**Supplementary Appendix**).

Analyses

The scheduled and final analyses were performed by an independent unblinded analytic team which was separate from the trial team. For this analysis, we used a minimally-informative prior for all model parameters. The analyses are described in the statistical appendix included with the **Supplementary Appendix**.

The adjusted odds ratio (aOR) for the primary outcome was calculated using a hierarchical Bayesian logistic regression model. This model incorporated data from all platform participants to adjust for age, country, temporal epoch (contiguous 26-week periods from the start of the domain), and eligibility and treatment assignment in other platform domains. Details of the model fitting and diagnostics are provided in the **Supplementary Appendix**. The treatment effect pertaining to cefazolin versus (flu)cloxacillin was calculated exclusively by using patients in this silo's randomized comparison (**Figure 1)** with pre-specified information sharing from the 87 pediatric participants enrolled in this silo (study ongoing). The probability of non-inferiority (pre-specified as an aOR<1.2; approximating an absolute margin of 2.5% absolute difference if mortality in the [flu]cloxacillin arm was 15%) and superiority (aOR<1.0) were estimated. The primary analysis for the primary outcome excluded participants with missing 90-day mortality data (complete case analysis). Post hoc analyses included analyses where all participants whose day 90 mortality data was missing were assumed to have survived or to have died.

Binary secondary outcomes were analyzed similarly. We planned a time-to-event analysis using a Bayesian Weibull proportional hazards model; however, this model's assumptions were not satisfied and so we present unadjusted Kaplan-Meier curves. Pre-specified subgroup analyses for the primary outcome were the presence or absence of endocarditis and the presence or absence of suspected/proven central nervous system (CNS) infection. During peer review, a post hoc subgroup analysis was conducted for a shorter and longer duration of pre-treatment (≤48 hours, 48-72 hours).

As analyses were conducted in a Bayesian framework, we report the means of the posterior distributions of treatment effects with 95% credible intervals and where appropriate associated posterior probabilities. No P-values were calculated, and no adjustment for multiple analyses was pre-specified. Therefore, for the secondary estimands, excluding safety outcomes, the 95% credible intervals should not be used to draw definitive conclusions.

Interim Analyses and Decision Rules

As a perpetual adaptive Bayesian platform, simulations were conducted to optimize trial operating characteristics such as number and timing of interim analyses, decision rule thresholds, and with a maximum ceiling sample size of 6,000 patients.[12] We planned and conducted scheduled analyses after every 500 platform patients reached day 90 with pre-specified stopping rules at each analysis for each domain.[12] For the MSSA silo of the backbone domain, if the posterior probability of non-inferiority for cefazolin was greater than 99% in adults, the independent data and safety monitoring committee (DSMC) could recommend stopping recruitment. If the posterior probability of superiority for adults was less than 1%, the DSMC could recommend stopping for futility. If no threshold was met, recruitment was to continue unless the DSMC had substantial safety concerns which were not prespecified. For the MSSA non-inferiority comparison, simulations indicated an approximate frequentist type I error of 2%.

On June 21, 2024, at the fourth planned interim analysis, the DSMC requested recruitment be paused in the MSSA silo because of a safety signal involving AKI. Subsequently, on August 5, 2024, they recommended closing the MSSA silo because the statistical trigger for non-inferiority had been met, and the safety signal persisted. On August 7, 2024, without additional knowledge on study outcomes, the trial steering committee accepted the recommendation.

Study Oversight

International coordination was provided by the University of Melbourne with the direction of a global trial steering committee.[11] Within each participating region, study coordination and monitoring was provided by a sponsor who was responsible under each region's laws and governance.

Results

Study Population

Of 8546 patients screened as of June 21, 2024, 2602 adults (30.4%) were recruited to the platform and underwent one or more randomizations (**Fig. S1**). Of those, 1757 had MSSA (67.5%) and 1341/1757 (76.3%) participants were randomized in the MSSA silo (671 to cefazolin, 670 to (flu)cloxacillin; **Figure 1**). Fifty-four of these patients (4.0%) were lost to follow-up, leaving 1287 patients in the primary outcome analysis. Recruitment to SNAP is ongoing and to maintain trial integrity, the results for the pediatric patients, MRSA silo, and other domains remain blinded.

Baseline characteristics are presented by treatment assignment in **Table 1.** The median age was 66 years (interquartile range [IQR] 53–76) and 421 (31.4%) were female, which are representative of a typical cohort with *S. aureus* bacteremia (**Tables S1-S3**). The most common site of infection was osteoarticular (32.1%) (**Table 2)**. Antibiotics received prior to randomization allocation are presented in **Table S4**. Four hundred and eighty-eight patients (36.4%) were receiving cefazolin at enrollment and 665 (49.6%) (flu)cloxacillin, with similar numbers across the assigned treatment groups. More than 99% of patients in this study received infectious diseases consultations. Both treatment groups received a similar duration of intravenous and oral antibiotics (**Table S5**).

Primary Outcome

Ninety-seven of 645 evaluable patients (15.0%) in the cefazolin group and 109 of 642 evaluable patients (17.0%) in the (flu)cloxacillin group met the primary outcome of mortality at 90 days (aOR 0.81; 95% CrI 0.59–1.12) corresponding to a posterior probability of non-inferiority of 99.2% and superiority of 89.8% (**Table 3, Fig. S2)**. In the 3.8% of patients with missing primary outcome data, the baseline characteristics were similar to those with complete data, apart from a younger age and history of injecting drug use in both treatment arms (**Table S6**). The complementary analyses accounting for missing outcomes were congruent with the primary analysis having posterior probabilities of non-inferiority exceeding 99%. A further post hoc sensitivity analysis using a broader prior demonstrated similar results to the primary analysis (**Table S7**).

Secondary Outcomes

Secondary outcomes are presented in **Table 3.** At every time point prior to day 90, cefazolin had a probability of non-inferiority for mortality exceeding 98.8% and a probability of superiority exceeding 94.1%. Unadjusted Kaplan-Meier plots for mortality are provided in **Figure S3**.

Patients receiving cefazolin experienced less AKI than those receiving (flu)cloxacillin (92/660 [13.9%] versus 127/648 [19.6%]; aOR 0.67; 95%CrI 0.50–0.89; probability of superiority 99.7%); including a 94.6% posterior probability of a reduced risk of initiating renal replacement therapy within 90 days (**Table 3, Fig. S4**). The staging of AKI is presented in the **Table S9**.

Protocol-adherent and Prespecified Subgroup Analyses

The protocol-adherent analysis demonstrated a 95.4% probability of cefazolin being non-inferior to (flu)cloxacillin (**Table 3**). Mortality for patients with endocarditis was 8/51 (15.7%) for cefazolin and 13/53 (24.5%) for (flu)cloxacillin (aOR 0.54; 95%CrI 0.20–1.43; probability of non-inferiority 94.4% and superiority 88.9%). There were only 7 patients with a central nervous system infection, precluding meaningful comparison.

Reported Adverse Events

Serious adverse reactions related to study drugs are detailed in **Table 3** and **Table S10.** Overall, cefazolin had substantially lower odds of a serious adverse reaction being reported (aOR 0.41; 95%CrI 0.21–0.77) and of having to be discontinued due to adverse events (aOR 0.21; 95%CrI 0.11–0.38).

Discussion

In this pragmatic randomized clinical trial in adults with MSSA bacteremia, cefazolin was non-inferior to (flu)cloxacillin for the primary outcome of all-cause mortality at 90 days. Cefazolin also had lower risk of acute kidney injury; initiation of renal replacement therapy; and of treatment discontinuation due to adverse events. These data also suggest that cefazolin may reduce mortality, with an estimated probability of superiority of 94–99% at days 14, 28, and 42 which remained approximately 90% by day 90.

The adjusted odds ratio for 90-day mortality of 0.81 (CrI 0.59–1.12) for cefazolin versus (flu)cloxacillin is similar to pooled estimates from observational studies of an overall odds ratio of 0.73 compared to the antistaphylococcal penicillins and 0.31 to 0.92 for individual drugs within the class.[9] While patients with *S. aureus* bacteremia are at direct risk of acute kidney injury by virtue of acute illness,[17] the potential nephrotoxicity of the antistaphylococcal penicillins has been recognized since the 1960s.[18] While we used flucloxacillin and cloxacillin in this trial, other antistaphylococcal penicillins have similar antibacterial activity and chemical structures,[19] and based on observational data are likely to confer a similar risk of

nephrotoxicity or treatment discontinuation for adverse events when compared to cefazolin.[9] Therefore, the mortality and nephrotoxicity findings reported here may apply to other antistaphylococcal penicillins other than (flu)cloxacillin.

Cefazolin has some potential disadvantages compared to (flu)cloxacillin, including higher drug acquisition costs in some markets, and a potentially broader antimicrobial spectrum. However, these concerns are balanced by its longer half-life, reduced toxicity, and possible superiority to (flu)cloxacillin.

Limitations

For reasons of practicality and feasibility, this was an open-label trial; however, the primary outcome, all-cause mortality at 90 days, was objective and the decision criteria to stop the trial was based on pre-specified statistical rules. We had limited control over physician decisions post-randomization. In the presence of persistent bacteremia or other clinical factors, physicians could have done more diagnostic testing, changed or added antibiotics, or performed more aggressive source control. We did not collect and therefore cannot present these data. We donot believe post-randomization care would have been different by knowledge of treatment assignment, but this is a limitation of the open-label design and also that variability of practices across sites may have introduced heterogeneity. Fortunately, a change of assigned therapy due to clinician perceived treatment failure was less than 2.5% and the results of the protocol-adherent population analysis were consistent with the primary analysis. Second, we have not yet tested isolates for the presence of the cefazolin inoculum effect and will report this microbiological subgroup when testing of all isolates has been performed at central reference laboratories. In the interim, concerns about the clinical relevance of the cefazolin inoculum effect[7] should be substantially allayed by the narrow non-inferiority margin and similar estimated treatment effect in patients with infective endocarditis compared to the overall population. Third, we presented a complete case analysis for the primary outcome (3.8% missing), with complementary analyses where all missing patients were considered either alive or dead yielding near identical results.

Conclusion

This pragmatic international, open-label platform trial demonstrated that cefazolin was non-inferior to (flu)cloxacillin for 90-day all-cause mortality in the treatment of adults with MSSA bacteremia. Cefazolin was associated with fewer serious adverse reactions, in particular less acute kidney injury.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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Figure legend:

Figure 1

Title: Enrolment, randomization, and follow-up.

Notes: MSSA denotes methicillin-susceptible *Staphyloccocus aureus*.

Table 1. Baseline Characteristics of Patients		
	No./Total No.(%)	
Variable	Cefazolin (N=671)	(Flu)cloxacillin (N=670)
Demographics		
Median age (IQR)	66 (53–76)	66 (52–77)
Female sex	215 (32.0)	206 (30.7)
Median weight in kg (IQR)	82.0 (69.0–96.8)	80.0 (68.0–95.0)
Comorbidities		
Diabetes mellitus	231 (34.4)	217 (32.4)
Chronic kidney disease[1,2]	96 (13.9)	93 (14.3)
Cirrhosis	34 (5.1)	30 (4.5)
Cancer requiring any surgical or medical therapy within 12 months[3]	95 (14.2)	76 (11.3)
Dementia	23 (3.4)	32 (4.8)
Iatrogenic immunosuppression[4]	105 (15.6)	84 (12.5)
Implanted intravascular prosthesis or endovascular device[5]	64 (9.5)	58 (8.7)
Previous infective endocarditis	11 (1.6)	10 (1.5)
Underlying cardiac abnormality predisposing to endocarditis[6]	50 (7.5)	42 (6.3)
Injection drug use within 6 months	31 (4.6)	52 (7.7)
Limitations to care[7]	80 (11.9)	84 (12.5)
Prognostic Factors		
Median hours to index blood culture positivity (IQR)	15.7 (12.0–21.0)	15.3 (12.0–20.7)
Median hours from index blood culture to platform entry (IQR)	52.3 (43.5–65.5)	50.9 (42.3–64.7)
Median hours from index blood culture to silo entry (IQR)	62.2 (50.2–67.9)	61.1 (50.0–67.5)
Intensive care unit at time of recruitment	74 (11.0)	72 (10.7)
Median Pitt bacteremia score (IQR)	0 (0–1)	0 (0–1)
Median C-reactive protein (mg/L) (IQR)	192.0 (113.5–266.5)	203.1 (122–283.8)

[1]Defined as estimated glomerular filtration rate below 50mL/min/1.72m^2; [2]Four patients (0.3%) were receiving baseline hemodialysis or peritoneal dialysis (protocol deviations). They were retained in the intention-to-treat population; [3]Excluding non-melanoma skin cancers; [4]Within past 3 months has received any of the following: prednisone >0.5mg/kg/day (or equivalent) for more than 14 days, cyclosporine, tacrolimus, or sirolimus, azathioprine, leflunomide, or mycophenolate, cyclosporine or methotrexate, monoclonal antibodies such as infliximab, cancer chemotherapy, bone marrow transplantation, other immunosuppressive agents equal to above, AIDS;[5]Not including central lines whether temporary or tunneled; [6]Excluding prosthesis above; [7]Active orders or advance care directives limiting resuscitation and intensive care admission.

Table 2. Focus of Infection		
	No./Total No.(%)	
Focus of Infection[1]	Cefazolin (N=671)	(Flu)cloxacillin (N=670)
Osteoarticular (including epidural abscess)	220 (32.8%)	210 (31.3%)
Septic arthritis and extra-axial osteomyelitis	167 (24.9%)	151 (22.5%)
Vertebral osteomyelitis and/or septic discitis	61 (9.1%)	59 (8.8%)
Epidural abscess	45 (6.7%)	49 (7.3%)
Orthopedic hardware-associated	34 (5.1%)	31 (4.6%)
Skin and soft tissue	177 (26.4%)	204 (30.4%)
Isolated skin and soft tissue	127 (18.9%)	136 (20.3%)
Intravascular catheter associated	103 (15.4%)	102 (15.2%)
Isolated intravascular catheter infection	91 (13.6%)	91 (13.6%)
Primary bacteremia (no other focus identified)	81 (12.1%)	80 (11.9%)
Endocarditis	55 (8.2%)	57 (8.5%)
Left-sided native valve	35 (5.2%)	33 (4.9%)
Right-sided native valve	13 (1.9%)	19 (2.8%)
Prosthetic valve	13 (1.9%)	9 (1.3%)
Pleuropulmonary	40 (6.0%)	42 (6.3%)
Other deep focus	27 (4.0%)	37 (5.5%)
Genitourinary	24 (3.6%)	22 (3.3%)
Native vascular or vascular graft	15 (2.2%)	16 (2.4%)
Unspecified	14 (2.1%)	13 (1.9%)
Cardiac or other implantable non-orthopedic device	12 (1.8%)	16 (2.4%)
Central Nervous System	5 (0.7%)	2 (0.3%)
[1]Patients may have had more than one focus of infection except where isolated line and skin and soft tissue infection is specified		

Table 3. Primary and Secondary Outcomes					
	No./Total No.(%)				
Outcome or Analysis	Cefazolin (N=671)	(Flu)cloxacillin (N=670)	Median Adjusted Odds Ratio (95%Credible Interval)	Posterior Probability of Cefazolin Non-Inferiority	Posterior Probability of Cefazolin Superiority
Primary Outcome - 90 Day Mortality					
Primary Analysis Population	97/645 (15.0%)	109/642 (17.0%)	0.81 (0.59–1.12)	99.2%	89.8%
Assuming all missing were dead	123/671 (18.3%)	137/670 (20.4%)	0.83 (0.63–1.10)	99.6%	90.7%
Assuming all missing were alive	94/671 (14.5%)	109/670 (16.3%)	0.81 (0.59–1.11)	99.3%	90.5%
Protocol-adherent Population	81/589 (13.8%)	73/520 (14.0%)	0.88 (0.61–1.26)	95.4%	76.0%
Secondary Outcomes					
Acute Kidney Injury within 14 days[1]	92/660 (13.9%)	127/648 (19.6%)	0.67 (0.50–0.89)	>99.9%	99.7%
Hepatotoxicity within 14 days[2]	75/574 (13.1%)	78/562 (13.9%)	0.96 (0.68–1.35)	90.0%	59.7%
Any serious adverse reaction related to study drug	12/671 (1.8%)	33/670 (4.9%)	0.41 (0.21–0.77)	99.9%	99.8%
Day 14 Mortality	25/667 (3.7%)	37/665 (5.6%)	0.67 (0.39–1.11)	NR	NR
Day 28 Mortality	47/665 (7.1%)	70/665 (10.5%)	0.61 (0.41–0.91)	NR	NR
Day 42 Mortality	63/663 (9.5%)	86/662 (13.0%)	0.66 (0.46–0.94)	NR	NR
Change of antibiotic due to perceived inefficacy	14/671 (2.1%)	16/670 (2.4%)	1.05 (0.51–2.10)	NR	NR
Microbiologic Treatment Failure After Day 14	18/587 (3.1%)	12/566 (2.1%)	1.41 (0.71–2.78)	NR	NR
Diagnosis of New Foci of Infection after Day 14	28/623 (4.5%)	30/609 (4.9%)	0.87 (0.53–1.45)	NR	NR
Change of antibiotic due to adverse event	11/671 (1.6%)	61/670 (9.1%)	0.21 (0.11–0.38)	NR	NR
New Renal Replacement Therapy within 90 days	17/668 (2.5%)	27/657 (4.1%)	0.61 (0.33–1.11)	NR	NR
Ongoing Renal Replacement Therapy at 90 days	4/662 (0.6%)	3/643 (0.5%)	1.00 (0.31–3.16)	NR	NR
C. difficile infection[3]	14/664 (2.1%)	10/661 (1.5%)	1.28 (0.60–2.66)	NR	NR
Intravenous catheter complication requiring removal[4]	22/668 (3.3%)	35/665 (5.0%)	0.68 (0.39–1.16)	NR	NR

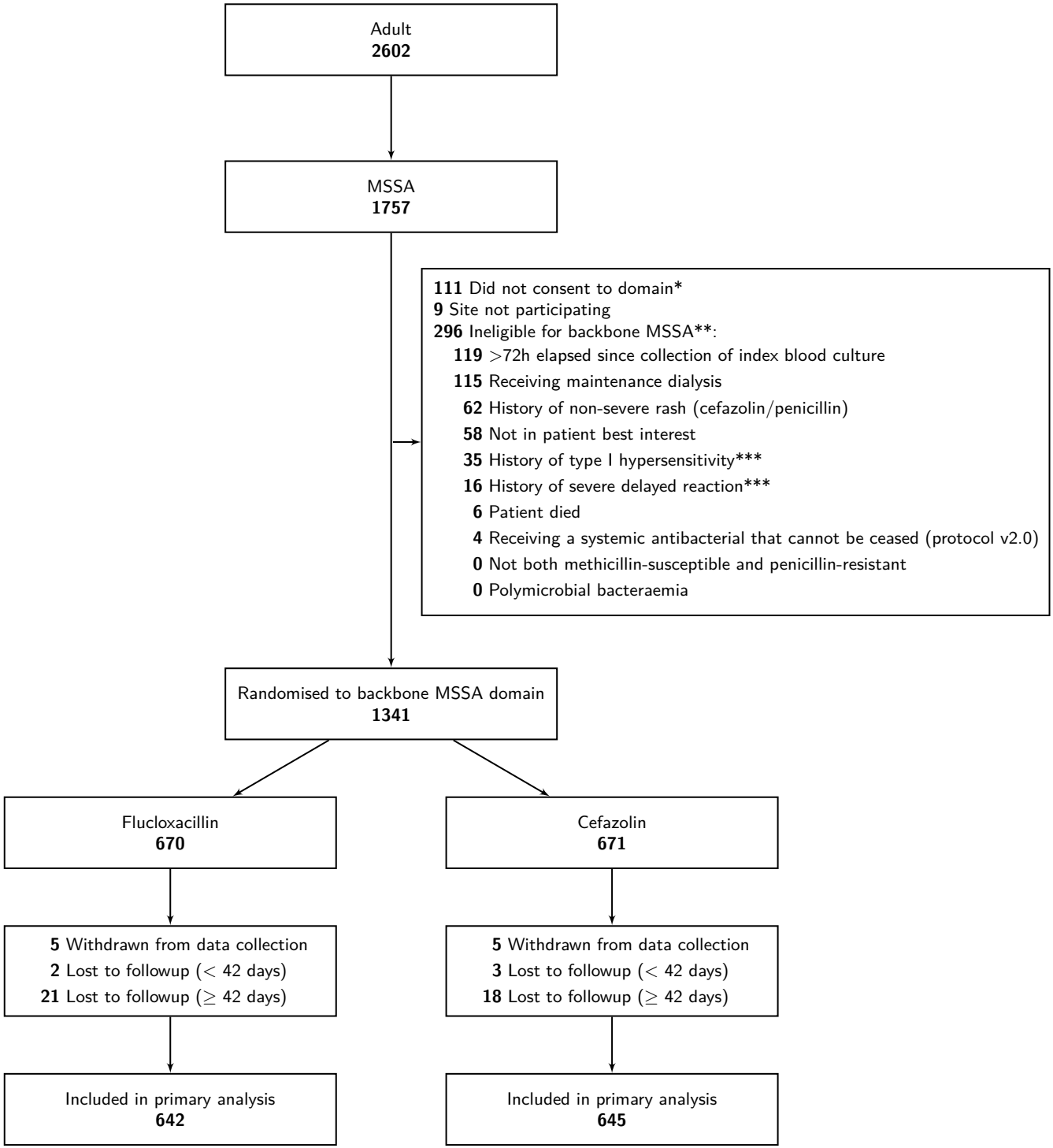
[1]Defined as a creatinine increase ⩾26.5 μmol/L (0.3mg/dL) or ⩾50% from baseline;

[2]Defined as alanine transaminase or gamma-glutamyl transferase increasing above 2.5 times the upper limit of normal;

[3]Defined as having symptoms, a positive test, and receiving therapy;

[4]While on assigned therapy.

[5]Posterior probabilities are not reported for secondary outcomes, with the exception of safety outcomes, and 95% credible intervals do not include adjustment for multiplicity.



* Includes consent not sought.
** Participants may meet more than one exclusion criteria.
*** To penicillin or cephalosporin.